Functions - Trigonometric Functions: Part 2

Course: Pre-Calculus-I

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Outline

1 The Tangent Function

2 Monotonicity and Extrema (Graph Reading)

3 The Pythagorean Identity and Other Relations

4 Applications and Practice

The Tangent Function

Definition

$$\boxed{\tan\theta = \frac{\sin\theta}{\cos\theta}} \qquad \text{wherever } \cos\theta \neq 0.$$

Where is it undefined? When $\cos\theta = 0$, i.e.

$$\theta = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

- **Domain:** $\mathbb{R} \setminus \left\{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\right\}$.
- **Range:** $\mathbb{R} = (-\infty, \infty)$.
- **Period:** π (not 2π).

Note

At $\theta = \pi/2 + k\pi$ the graph has a vertical asymptote.

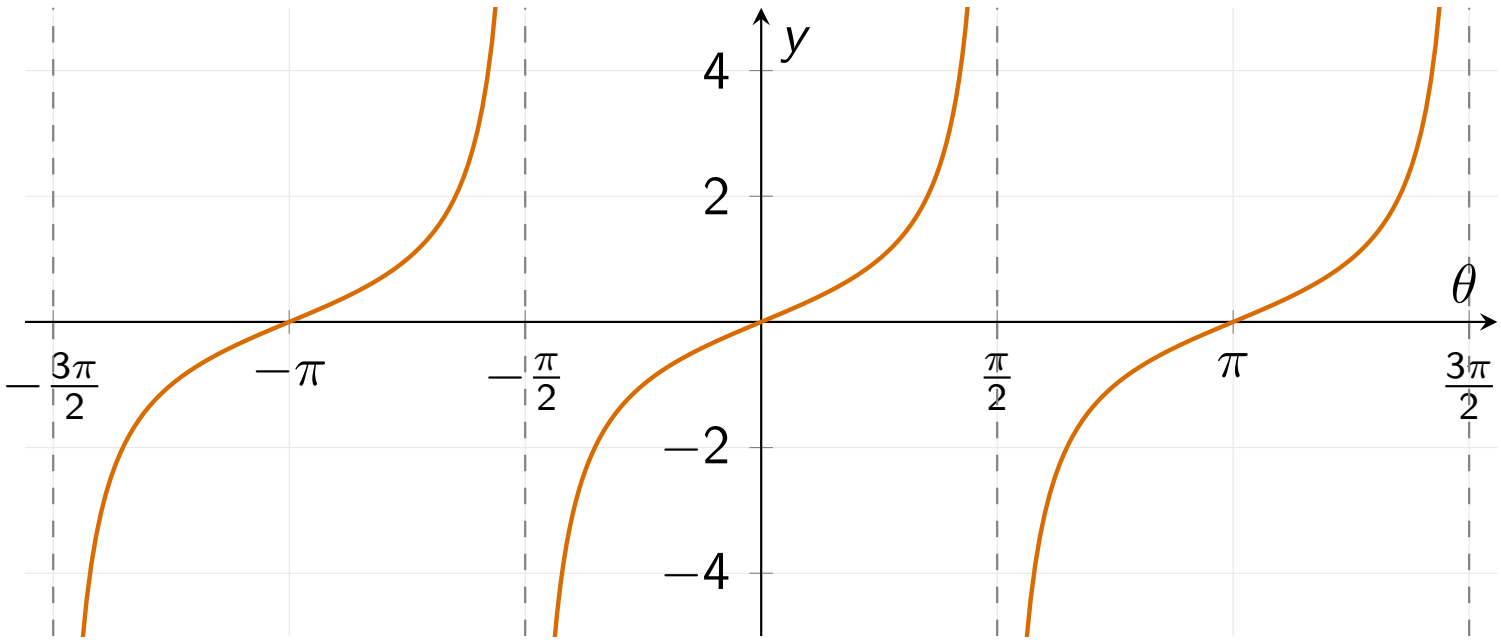
Values of $\tan\theta$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Observations:

- $\tan\theta$ increases without bound as $\theta \to \pi/2^{-}$.
- $\tan\theta$ decreases from $-\infty$ up to $\pi/2^{+}$.
- The graph repeats every π.

Graph of $f(\theta) = \tan\theta$



Vertical asymptotes at $\theta = \frac{\pi}{2} + k\pi$. Each branch increases from $-\infty$ to $+\infty$.

Increasing and Decreasing (Graph Reading)

Function	Increasing on	Decreasing on
$\sin\theta$ (one period $[0, 2\pi]$)	$[0, \frac{\pi}{2}] \cup [\frac{3\pi}{2}, 2\pi]$	$[\frac{\pi}{2}, \frac{3\pi}{2}]$
$\cos\theta$ (one period $[0, 2\pi]$)	$[\pi, 2\pi]$	$[0, \pi]$
$\tan\theta$ (one branch)	$(-\frac{\pi}{2}, \frac{\pi}{2})$	never

How to read this from the graph:

- Trace the curve left to right.
- Where the curve rises $\Rightarrow$ increasing.
- Where the curve falls $\Rightarrow$ decreasing.
- sin and cos alternate; tan always rises between asymptotes.

Note

The pattern repeats every 2π *for* sin *and* cos, *every* π *for* tan.

Maximum and Minimum Values (No Calculus)

Bounded Ranges

$$-1 \leq \sin\theta \leq 1, \qquad -1 \leq \cos\theta \leq 1.$$

So sin and cos have absolute maximum 1 and absolute minimum -1 on $\mathbb{R}$.

Transformed Functions

If $f(\theta) = A\sin(B\theta + C) + D$ (or with cos), then

$$D - |A| \leq f(\theta) \leq D + |A|.$$

Tangent

$\tan\theta$ has range $\mathbb{R}$ — *no* maximum or minimum on any interval that crosses an asymptote.

Example: $f(\theta) = 3\sin(2\theta) + 1$

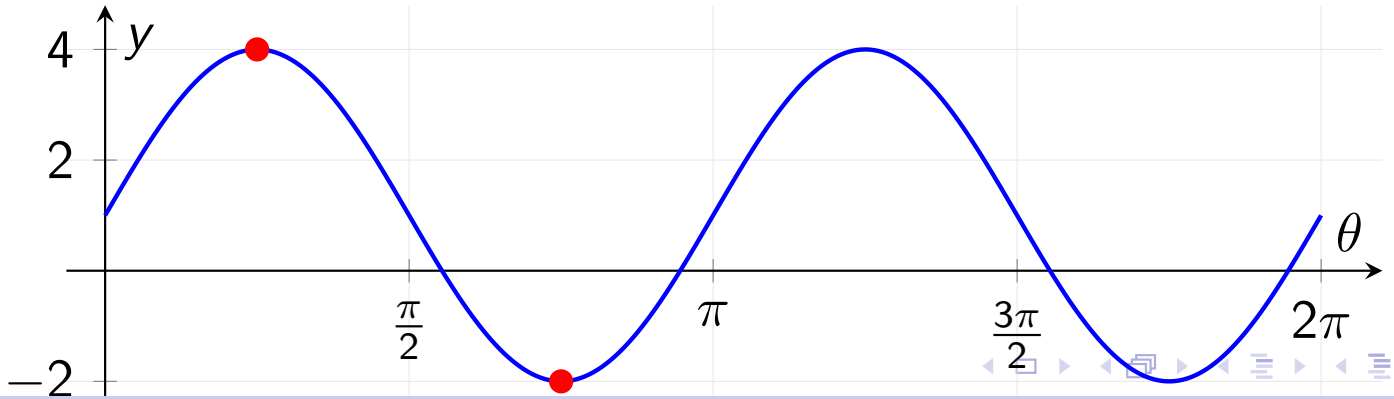
Identify: $A = 3,\ B = 2,\ D = 1.$

- Amplitude: 3, period: π, vertical shift: 1.
- Range: $[1-3,\ 1+3] = [-2,\ 4]$.
- Absolute maximum $= 4$, when $\sin(2\theta) = 1$, i.e.

$$2\theta = \frac{\pi}{2} + 2k\pi \ \Rightarrow\ \theta = \frac{\pi}{4} + k\pi.$$

- Absolute minimum $= -2$, when $\sin(2\theta) = -1$, i.e.

$$\theta = \frac{3\pi}{4} + k\pi.$$



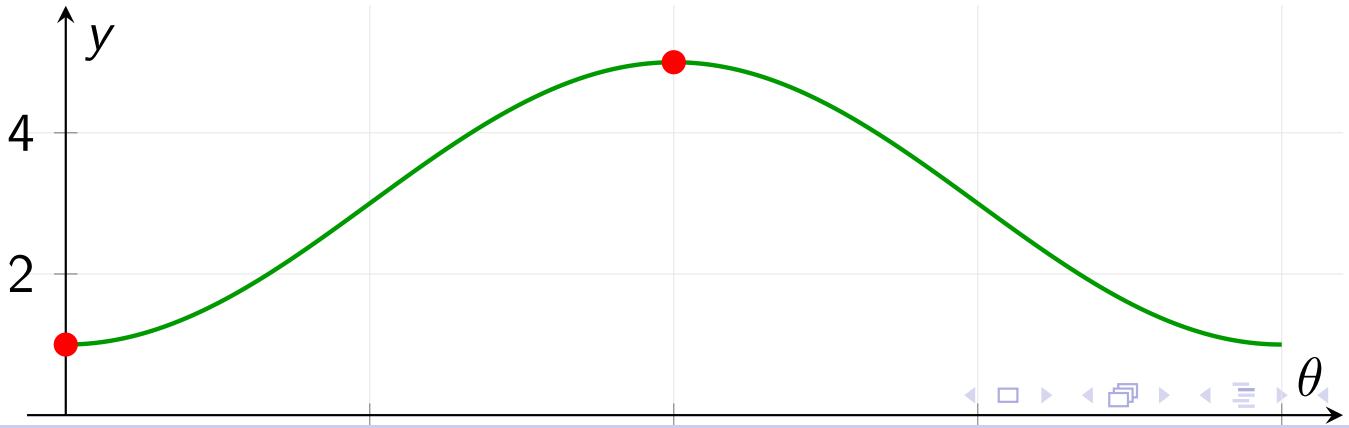
Example: $g(\theta) = -2\cos(\theta) + 3$

Identify: $A = -2,\ B = 1,\ D = 3.$

- Amplitude: $|A| = 2.$
- Period: $2\pi.$
- Vertical shift: $+3.$
- Range: $[3-2,\ 3+2] = [1,\ 5].$

Extrema:

- Maximum 5: occurs when $-2\cos\theta = 2$, i.e. $\cos\theta = -1$, $\theta = \pi + 2k\pi.$
- Minimum 1: occurs when $-2\cos\theta = -2$, i.e. $\cos\theta = 1$, $\theta = 2k\pi.$



Example: Evaluating at Special Angles

Compute:

1. $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$
2. $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$
3. $\tan\left(\frac{\pi}{4}\right) = 1$
4. $\sin\left(\frac{3\pi}{2}\right) = -1$
5. $\cos(\pi) = -1$

Compute expressions:

- $2\sin\left(\frac{\pi}{2}\right) + 3\cos(0) = 2(1) + 3(1) = 5.$
- $\sin^2\left(\frac{\pi}{4}\right) + \cos^2\left(\frac{\pi}{4}\right) = \frac{1}{2} + \frac{1}{2} = 1.$

Note

The Fundamental Identity

Since $P = (\cos\theta, \sin\theta)$ lies on $x^2 + y^2 = 1$:

$$\boxed{\cos^2\theta + \sin^2\theta = 1}$$

for *every* real θ.

Consequences:

- Divide by $\cos^2\theta$: $\quad 1 + \tan^2\theta = \sec^2\theta.$
- Divide by $\sin^2\theta$: $\quad \cot^2\theta + 1 = \csc^2\theta.$

Example: If $\sin\theta = \dfrac{3}{5}$ and θ is in the first quadrant, find $\cos\theta$ and $\tan\theta$.

$$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25} \Rightarrow \cos\theta = \frac{4}{5}, \qquad \tan\theta = \frac{3/5}{4/5} = \frac{3}{4}.$$

Other Useful Identities (Summary)

Identity	**Statement**
Pythagorean	$\sin^2\theta + \cos^2\theta = 1$
Quotient	$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$
Even / odd	$\sin(-\theta) = -\sin\theta, \quad \cos(-\theta) = \cos\theta$
Periodicity	$\sin(\theta + 2\pi) = \sin\theta,\ \cos(\theta + 2\pi) = \cos\theta$
Co-function	$\sin\left(\dfrac{\pi}{2} - \theta\right) = \cos\theta$
Double angle (sin)	$\sin(2\theta) = 2\sin\theta\cos\theta$
Double angle (cos)	$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

Applications of Trigonometric Functions

- **Periodic phenomena:**
  - Tides: height $h(t) = A\sin(Bt + C) + D$.
  - Temperature over a day: sinusoidal model.
  - Alternating current: $V(t) = V_0 \sin(\omega t)$.
- **Circular motion:** position $(x, y) = (R\cos\omega t,\ R\sin\omega t)$.
- **Right-triangle geometry:**

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

- **Waves and sound:** frequency, amplitude and phase.

Practice Problems

1. Convert: (a) 150° to radians; (b) $\dfrac{5\pi}{6}$ rad to degrees.
2. Using the unit circle, find $\sin\theta$, $\cos\theta$, $\tan\theta$ at $\theta = \dfrac{\pi}{6},\ \dfrac{\pi}{3},\ \dfrac{5\pi}{4}$.
3. State the domain and range of: (a) $y = \sin x$; (b) $y = \cos x$; (c) $y = \tan x$.
4. For $f(x) = 4\sin(3x) - 2$: state amplitude, period, range, and the values of the maximum and minimum.
5. For $g(x) = -3\cos(x) + 1$: state amplitude, period, range, and where the maximum/minimum occur.
6. Sketch one period of $f(x) = 2\sin\left(x - \dfrac{\pi}{4}\right)$ and label the max/min points.
7. Use $\sin^2\theta + \cos^2\theta = 1$ to find $\cos\theta$ if $\sin\theta = \dfrac{4}{5}$ and θ is in the first quadrant.
8. Simplify: $\dfrac{\sin\theta}{\cos\theta} \cdot \cos\theta + \sin^2\theta$.

Exercise: Chapter 4, Topic 4.1 — trigonometric functions, identities and graphs

Key Takeaways

- **Unit-circle definition:** $P = (\cos\theta, \sin\theta)$ on $x^2 + y^2 = 1$.
- **Domains and ranges:**
  - $\sin\theta$, $\cos\theta$: domain $\mathbb{R}$, range $[-1, 1]$.
  - $\tan\theta$: domain $\mathbb{R} \setminus \{\pi/2 + k\pi\}$, range $\mathbb{R}$.
- **Amplitude, period, shift:** for $A\sin(B\theta + C) + D$, amplitude $|A|$, period $2\pi/|B|$, range $[D - |A|, D + |A|]$.
- **Monotonicity** is read from the graph (rise/fall) and repeats each period.
- **Extrema** come from bounded ranges: $\sin, \cos \in [-1, 1]$, so max and min are read off immediately.
- **Fundamental identity:** $\sin^2\theta + \cos^2\theta = 1$.

Next: Maximum and Minimum Values